

Does Study Time Predict Test Performance Among Doral Academy Students?

I. Introduction

Academic performance is one of the most important concerns for high school students, teachers, and families. Students are often told to study more, but the specific relationship between the number of hours spent studying and the score earned on a test is not always clear. Some students believe that studying for many hours leads to higher scores, while others believe that the quality of studying matters more than the amount of time spent. Because students at Doral Academy High School take many major exams throughout the school year, this topic is especially relevant to our school community.

This project investigates whether the amount of time students spend preparing for a major test predicts their test performance. Our goal was not only to compare students who study more with students who study less, but also to examine whether the relationship between study hours and test scores appears positive and linear. Understanding this relationship can help students make better decisions about how much time to set aside before important assessments.

Previous research suggests that study habits and time investment can affect academic outcomes. Research on deliberate practice emphasizes that structured, focused preparation is more effective than unplanned cramming. Other studies have found that study habits are associated with exam outcomes and grade point average. However, much of the available research focuses on college students, leaving a gap in research involving high school students in selective or academically competitive school environments. Our study focuses on students at Doral Academy High School and their short-term performance on a major test.

The practical purpose of this study is to determine whether a simple, measurable variable - study time in hours - can be used to predict a student's test score. If a meaningful relationship exists, then students and teachers may be able to use this information to encourage more intentional study plans before major assessments.

II. Statistical Question

Our main statistical question is: Does the number of hours students spend studying for a major test significantly predict their test score among students at Doral Academy High School?

The explanatory variable in this study is study time, measured as the number of self-reported hours a student spent preparing for a major test. The response variable is test score, measured as the percentage score earned on that test. The population of interest is all students at Doral Academy High School in grades 9 through 12.

Because the study includes multiple forms of analysis, we used more than one statistical approach. First, we used a one-sample t-test to determine whether the mean number of study hours differed from a five-hour benchmark. Next, we used a two-sample t-test to compare students who studied fewer than five hours with students who studied five or more hours. Finally, we used linear regression to examine whether study time predicts test score.

Hypotheses for the one-sample t-test:

$H_0: \mu = 5 \text{ hours}$

$H_a: \mu \neq 5 \text{ hours}$

Hypotheses for the two-sample t-test:

$H_0: \mu_{\text{high}} - \mu_{\text{low}} = 0$

$H_a: \mu_{\text{high}} - \mu_{\text{low}} > 0$

Where μ_{high} is the true mean test score for students who studied five or more hours, and μ_{low} is the true mean test score for students who studied fewer than five hours.

III. Data Collection

To collect our data, we surveyed 50 Doral Academy students from grades 9 through 12. Students were recruited during study hall periods and asked whether they were willing to participate in an anonymous academic survey. Students who agreed were given a link to an online form and completed the survey on their phones. The survey asked students to report the number of hours they studied for their most recent major test and the score they earned on that test.

Our intended sampling design was stratified random sampling by grade level. We wanted each grade level to be represented in the sample so the results would not be dominated by only one class year. With a total sample size of 50 students, this meant that approximately 12 to 13 students were included from each grade level. This design was chosen because study habits may vary by grade, especially since upperclassmen may have more experience preparing for major exams.

The survey also included demographic information such as grade and gender. This information was not the main focus of the analysis, but it helped us understand the makeup of the sample. The main variables used in the analysis were study hours and test score. Study hours were measured on a scale from 0 to 12 hours, and test scores were measured from 0 percent to 100 percent.

Because the data were self-reported, we recognize that some response bias may be present. Students may not remember the exact number of hours they studied, and some students may overestimate their study time or test score. However, the survey was anonymous, which was intended to reduce pressure and encourage honest responses.

Operational Definitions

X variable: Study time, measured in self-reported hours spent preparing for a major test.

Y variable: Test score, measured as the percentage score earned on the major test.

Sample size: $n = 50$ students.

Sampling frame: Doral Academy students in grades 9-12.

IV. Data Display

The descriptive statistics show that the average reported study time was 4.58 hours, with a standard deviation of 2.06 hours. The five-number summary for study hours was minimum = 1, Q1 = 3, median = 5, Q3 = 7, and maximum = 8. This indicates that most students in the sample studied between 3 and 7 hours for their major test.

The mean test score was 83.24 percent, with a standard deviation of 5.62 percentage points. The scores suggest that most students in the sample performed fairly well overall, but there was still enough variation to analyze whether study time helped explain differences in performance.

Variable	n	Mean	SD	Min	Q1	Median	Q3	Max
Study Hours	50	4.58	2.06	1	3	5	7	8
Test Score (%)	50	83.24	5.62	--	--	--	--	--

Figure 1. Summary statistics for study hours and test scores in the sample of 50 Doral Academy students.

Based on the distribution of study hours, the five-hour value was a reasonable benchmark to use for comparison. The median study time was exactly five hours, and the mean study time rounded to approximately five hours. This allowed us to divide students into a lower-study group and a higher-study group for the two-sample comparison.

For a stronger final presentation, this section should include a clean table of the raw survey data or a smaller selected data table, along with a graph such as a scatterplot of study hours versus test score. The winning project example used clear data tables and labeled figures, so we followed that structure by separating the data display from the statistical analysis.

V. Data Analysis

One-Sample t-Test for Mean Study Time

The first inference procedure tested whether the average number of hours Doral Academy students studied differed from five hours. The hypotheses were $H_0: \mu = 5$ hours and $H_a: \mu \neq 5$ hours. The sample mean was 4.58 hours, and the p-value was 0.155.

Because the p-value of 0.155 is greater than the significance level of 0.05, we fail to reject the null hypothesis. There is not convincing statistical evidence that the average number of hours Doral Academy students study differs from five hours. In context, the sample mean of 4.58 hours rounds close to the five-hour benchmark, so the typical student in the sample appears to study about five hours for a major test.

Conditions for the one-sample t-test:

Random: Students were selected using a stratified approach by grade level to improve representation across grades. This condition is reasonably met.

Independence: The sample of 50 students is less than 10 percent of all students at Doral Academy High School, so the 10 percent condition is met.

Normal/Large Sample: Since $n = 50$, the sample size is large enough for the sampling distribution of the mean to be approximately normal by the Central Limit Theorem.

Since the conditions were reasonably met, the one-sample t-test was appropriate for this part of the analysis.

V. Data Analysis, Continued

Two-Sample t-Test for Difference in Mean Test Scores

After establishing that five hours was a reasonable study-time benchmark, we divided students into two groups: students who studied fewer than five hours and students who studied five or more hours. We then compared the mean test scores of the two groups.

The hypotheses for this test were $H_0: \mu_{\text{high}} - \mu_{\text{low}} = 0$ and $H_a: \mu_{\text{high}} - \mu_{\text{low}} > 0$. The alternative hypothesis was one-sided because we wanted to test whether students who studied five or more hours scored higher, on average, than students who studied fewer than five hours.

The p-value for the two-sample t-test was 0.00000000000046, which is much smaller than the significance level of 0.05. Therefore, we reject the null hypothesis. There is convincing statistical evidence that students who studied five or more hours scored higher on tests than students who studied fewer than five hours.

In context, this result suggests that the five-hour benchmark is not only a description of the typical study time but also a meaningful dividing point between lower and higher test performance in our sample. Students who reached or exceeded five hours of study tended to perform better than students who studied less than five hours.

Conditions for the two-sample t-test:

Random: The original sample was designed to include students from multiple grade levels, and students were placed into the comparison groups based on their reported study time.

Independence: Individual student responses were independent, and the sample size was less than 10 percent of the population of Doral Academy students.

Normal/Large Sample: The total sample size was 50. Because both groups came from the same survey sample and contained multiple students, the t-procedure was reasonable for comparing group means.

V. Data Analysis, Continued

Regression Analysis

To examine whether study time predicts test score, we used a linear regression model with study hours as the explanatory variable and test score as the response variable. The regression equation was: Predicted score = $71.501 + 2.67(\text{Study Hours})$.

The slope of 2.67 means that for each additional hour a student studied, the predicted test score increased by about 2.67 percentage points. This interpretation is important because it gives the relationship a practical meaning. For example, a student who studies two additional hours would be predicted to score about 5.34 percentage points higher, assuming the linear model is appropriate.

The coefficient of determination was $R\text{-squared} = 0.69$. This means that about 69 percent of the variation in test scores in our sample can be explained by the number of hours students studied. This is a fairly strong result for a study involving human behavior, since academic performance can also be affected by many other factors such as sleep, motivation, prior knowledge, attendance, and test difficulty.

The correlation was reported as $r = 0.68$, indicating a positive association between study time and test score. Since the slope is positive, the model supports the idea that students who studied more tended to earn higher scores. The relationship was described as strong, positive, and approximately linear.

Figure 2. A scatterplot of study hours versus test scores should be included here in the final visual version. The graph should show study hours on the x-axis and test score percentage on the y-axis, along with the least-squares regression line.

VI. Conclusion

The results of this study support the conclusion that study time predicts test performance among Doral Academy students in our sample. The one-sample t-test showed that the average study time was not significantly different from the five-hour benchmark. This helped justify using five hours as a meaningful cutoff point for comparing students with lower and higher study time.

The two-sample t-test provided convincing evidence that students who studied five or more hours scored higher, on average, than students who studied fewer than five hours. This result directly answers an important practical question: students who spent more time preparing for a major test tended to perform better.

The regression analysis also supported the same overall conclusion. The positive regression slope showed that each additional hour of study was associated with an increase of about 2.67 percentage points in predicted test score. The R-squared value of 0.69 indicated that study time explained a substantial portion of the variation in test scores. Together, these results suggest that increased and structured studying can have a meaningful positive impact on academic performance.

However, the study does not prove that study time alone causes higher scores. Since this was a survey rather than a controlled experiment, other variables may also explain part of the relationship. Students who study more may also be more motivated, more organized, or more attentive in class. Still, the evidence from our sample strongly suggests that study time is an important predictor of test performance.

VII. Reflection

This project helped us understand how statistical inference can be used to answer a real question in our school community. We began with a common academic concern - whether studying more actually leads to better test scores - and used data to investigate it. By applying a one-sample t-test, a two-sample t-test, and regression analysis, we were able to examine the question from several perspectives.

One strength of our study was that the topic was relevant and easy for students to understand. Study time and test scores are variables that most high school students can relate to. Another strength was that the sample included students from grades 9 through 12, which made the sample more representative than if we had only surveyed one grade level.

One limitation is that the data were self-reported. Some students may not have remembered their exact study time, and others may have rounded their answer. Test scores were also self-reported, which could introduce bias if students were uncomfortable sharing low scores. Although the survey was anonymous, response bias may still have affected the accuracy of the data.

Another limitation is that test difficulty was not controlled. Students may have been reporting scores from different subjects, different teachers, or different types of major tests. A score of 85 percent on one exam may not represent the same level of performance as an 85 percent on another exam. This makes it harder to compare all students as if they took the same assessment.

Motivation is also a possible confounding variable. A student who studies more may also be the type of student who pays closer attention in class, completes homework consistently, sleeps more before tests, or cares more about grades. Therefore, the higher scores may not be caused by study time alone. Future research should attempt to measure these additional factors.

VIII. Implications and Future Research

The results of this study suggest that students should be encouraged to create structured study plans before major tests. Since students who studied five or more hours performed better, teachers and counselors may consider recommending that students begin preparing several days before a major assessment instead of relying on last-minute cramming.

Doral Academy could also offer workshops on evidence-based study strategies. These workshops could teach students how to use active recall, practice tests, spaced repetition, note review, and distraction-free study sessions. The number of hours studied matters, but the quality of those hours may matter just as much or even more.

A reasonable recommendation based on this study is that students should aim for at least five to six hours of preparation before a major test, especially for challenging subjects. This does not mean that every student needs the exact same amount of time, but it provides a useful guideline. Students who currently study fewer than five hours may benefit from increasing their preparation time.

Future research could improve this study by using a larger and more diverse sample from multiple schools. Researchers could also collect data over an entire school year instead of relying on one recent test. A longitudinal study would make it possible to see whether changes in study habits lead to changes in academic performance over time.

Future studies should also examine the quality of studying. For example, students could report whether they used active recall, flashcards, practice problems, group studying, tutoring, or rereading notes. Researchers could then compare which study strategies are most strongly associated with higher test scores. Other variables such as sleep, stress, attendance, and motivation should also be measured so their effects can be separated from the effect of study time.

IX. Citations

Crede, M., & Kuncel, N. R. (2008). Study habits, skills, and attitudes: The third pillar supporting collegiate academic performance. *Perspectives on Psychological Science*.

Nonis, S. A., & Hudson, G. I. (2010). Performance of college students: Impact of study time and study habits. *Journal of Education for Business*.

Plant, E. A., Ericsson, K. A., Hill, L., & Asberg, K. (2005). Why study time does not predict grade point average across college students: Implications of deliberate practice for academic performance. *Contemporary Educational Psychology*.